

Functional Analysis, PDE, and Finite Element Methods

Reading notes on *Finite Element Methods for Maxwell's Equations* by Peter Monk

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1 Introduction

This note covers Chapter 1,2,3,4,6 of the book *Finite Element Methods for Maxwell's Equations* by Peter Monk. I found some places ambiguous so I write it in a clearer way. I assume readers know fundamental functional analysis and PDE. At least readers should know Hilbert Spaces, Banach spaces, continuous linear maps, and Riesz Representation Theorem, which will be the starting point of this note. One should also know Sobolev Spaces. Chapter 5 of Evans' PDE is sufficient. Since I have a background in computational elasticity, I will add some examples in solid mechanics as well.

2 Starting from Riesz Representation Theorem

We begin by stating the Riesz representation theorem without proof.

Throughout these notes, the inner product is linear in its first argument. We use the same convention for sesquilinear forms: they are linear in the first argument and conjugate-linear in the second argument.

Theorem 1 – Riesz Representation Theorem

Let H be a real or complex Hilbert space. For every bounded linear functional $f \in H'$, there exists a unique $y_f \in H$ such that

$$(x, y_f) = f(x), \quad \forall x \in H.$$

Moreover,

$$\|f\|_{H'} = \|y_f\|_H.$$

Thus the map $y \mapsto (\cdot, y)$ identifies H isometrically with its continuous dual H' ; in the complex case, this identification is conjugate-linear.

Remark 2 – The position of the Riesz representative

The representative y_f must appear in the second argument of the inner product:

$$f(x) = (x, y_f).$$

Indeed, under our convention the inner product is linear in its first argument and conjugate-linear in its second argument. Therefore, $x \mapsto (x, y_f)$ is linear, as required for f . If one instead wrote (y_f, x) , the resulting map would be conjugate-linear in x in the complex case and hence could not represent a general linear functional. For real Hilbert spaces, the distinction disappears.

Next, let us state and prove a more interesting result - Lax-Milgram Lemma - which will be used in finite element analysis very frequently. You can say it is a non-symmetric analogue of Riesz Representation theorem. But before that, let us define a useful term which will appear in the note all the time.

Definition 3 – Coercivity

Let H be a real or complex Hilbert space. A sesquilinear form $a : H \times H \rightarrow \mathbb{C}$ is called

coercive if there exists a constant $\alpha > 0$ such that

$$|a(v, v)| \geq \alpha \|v\|_H^2, \quad \forall v \in H.$$

Lemma 4 – Lax-Milgram Lemma

Let H be a real or complex Hilbert space, and let $a : H \times H \rightarrow \mathbb{C}$ be a bounded sesquilinear form. Suppose a is bounded, i.e., there exists a constant $M > 0$ such that

$$|a(u, v)| \leq M \|u\|_H \|v\|_H, \quad \forall u, v \in H,$$

and is coercive, i.e., there exists another constant $\alpha > 0$

$$|a(v, v)| \geq \alpha \|v\|_H^2, \quad \forall v \in H.$$

Then, for every bounded linear functional $f \in H'$, there exists a unique $u \in H$ such that

$$a(u, v) = \overline{f(v)}, \quad \forall v \in H.$$

Moreover, the solution satisfies the estimate

$$\|u\|_H \leq \frac{1}{\alpha} \|f\|_{H'}.$$

Let us parse what it talks about. First, it tells us that if a is bounded and coercive, then the equation $a(u, v) = \overline{f(v)}$ has a unique solution u . Second, it tells us that this unique solution is stable: if the norm of f changes a little bit, the change of the solution u will be bounded. Let us see the proof first, and then see a simple example in PDE.

Proof. **Step 1: Define the operator $A : H \rightarrow H$.** For each fixed $u \in H$, the map $v \mapsto \overline{a(u, v)}$ is a bounded linear functional on H . By the Riesz representation theorem, there exists a unique $Au \in H$ such that

$$(Au, v) = a(u, v), \quad \forall v \in H.$$

This defines a bounded linear operator $A : H \rightarrow H$.

Step 2: Rewrite the variational problem as an operator equation. By the Riesz representation theorem, there exists a unique $w \in H$ such that

$$(v, w) = f(v), \quad \forall v \in H.$$

By conjugate symmetry, $(w, v) = \overline{f(v)}$. Thus the equation $a(u, v) = \overline{f(v)}$ for every $v \in H$ is equivalent to

$$Au = w.$$

It remains to prove that A is bijective.

Step 3: Prove injectivity and closedness of the range. For every $u \in H$, coercivity and the Cauchy–Schwarz inequality give

$$\alpha \|u\|_H^2 \leq |a(u, u)| = |(Au, u)| \leq \|Au\|_H \|u\|_H.$$

Hence

$$\|Au\|_H \geq \alpha \|u\|_H.$$

In particular, A is injective, and the same lower bound implies that $\text{Ran } A$ is closed in H . Indeed, let $Au_n \in \text{Ran } A$ be a sequence such that $Au_n \rightarrow y$ in H . The lower bound gives

$$\|u_n - u_m\|_H \leq \frac{1}{\alpha} \|Au_n - Au_m\|_H.$$

Thus (u_n) is a Cauchy sequence in H . Since H is complete, there exists $u \in H$ such that $u_n \rightarrow u$. The continuity of A then gives $Au_n \rightarrow Au$. Since $Au_n \rightarrow y$ as well, we have $y = Au \in \text{Ran } A$. Therefore, $\text{Ran } A$ is closed.

Step 4: Prove surjectivity. Let $v \in (\text{Ran } A)^\perp$. Then

$$a(u, v) = (Au, v) = 0, \quad \forall u \in H.$$

Taking $u = v$ and using coercivity gives $v = 0$. Therefore $(\text{Ran } A)^\perp = \{0\}$, so $\text{Ran } A$ is dense in H . Since the range is both closed and dense, $\text{Ran } A = H$, and A is surjective.

Step 5: Conclude existence, uniqueness, and stability. The equation $Au = w$ therefore has a unique solution $u \in H$. Moreover,

$$\alpha \|u\|_H \leq \|Au\|_H = \|w\|_H = \|f\|_{H'},$$

which yields

$$\|u\|_H \leq \frac{1}{\alpha} \|f\|_{H'}.$$

■

The importance of the Lax–Milgram lemma is that it gives existence, uniqueness, and stability of a weak solution all at once. Existence means that every admissible right-hand side f produces a solution u ; uniqueness means that the weak formulation determines only one such u ; and stability means that the solution depends continuously on the data. More precisely, the estimate

$$\|u\|_H \leq \frac{1}{\alpha} \|f\|_{H'}$$

shows that a small perturbation of f can cause only a controlled perturbation of u . This is especially important in numerical analysis, because data, quadrature, and discretization inevitably introduce small errors on f . In practice, Lax–Milgram therefore reduces the analysis of many linear elliptic PDEs to identifying the correct Hilbert space and verifying boundedness and coercivity of the weak form. The following basic example illustrates this principle.

Example 5 – Poisson equation with homogeneous Dirichlet boundary data

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $f \in L^2(\Omega)$. Consider

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Multiplying the equation by a test function $v \in H_0^1(\Omega)$ and integrating by parts gives the weak formulation: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = F(v), \quad \forall v \in H_0^1(\Omega),$$

where

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad F(v) = \int_{\Omega} f v \, dx.$$

Set $H = H_0^1(\Omega)$ and use the energy norm $\|v\|_H = \|\nabla v\|_{L^2(\Omega)}$. By the Cauchy–Schwarz inequality,

$$|a(u, v)| \leq \|u\|_H \|v\|_H,$$

so a is bounded. Moreover,

$$a(v, v) = \int_{\Omega} |\nabla v|^2 \, dx = \|v\|_H^2,$$

so a is coercive with coercivity constant $\alpha = 1$. The Poincaré inequality shows that F is a bounded linear functional on H . Therefore, the Lax–Milgram lemma gives a unique weak solution $u \in H_0^1(\Omega)$, together with the stability estimate

$$\|u\|_H \leq \|F\|_{H'} \leq C \|f\|_{L^2(\Omega)}.$$

Thus existence, uniqueness, and stability of the Poisson problem follow directly from boundedness and coercivity, without requiring an explicit formula for u .

Lax–Milgram is powerful, but it is not universal. In particular, after a second-order equation is rewritten as a first-order system with several unknowns, the resulting bilinear form may fail to be coercive on the whole product space. The Poisson equation itself already provides a simple example.

Example 6 – Mixed Poisson problem: failure of coercivity

Consider again

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

but now introduce the flux $\sigma = \nabla u$. The equation becomes the first-order system

$$\begin{cases} \sigma - \nabla u = 0 & \text{in } \Omega, \\ -\operatorname{div} \sigma = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The following derivation is initially only formal. Assume temporarily that u , σ , and the test functions are smooth; we will determine their natural Sobolev spaces only after obtaining the integral identities. Multiply $\sigma - \nabla u = 0$ by an arbitrary smooth *vector-valued* test function τ and integrate over Ω :

$$\int_{\Omega} \sigma \cdot \tau \, dx - \int_{\Omega} \nabla u \cdot \tau \, dx = 0.$$

Green's formula gives

$$\int_{\Omega} \nabla u \cdot \tau \, dx = - \int_{\Omega} u \operatorname{div} \tau \, dx + \int_{\partial\Omega} u (\tau \cdot n) \, ds.$$

Because $u = 0$ on $\partial\Omega$, the boundary integral vanishes. Hence we formally obtain

$$\int_{\Omega} \sigma \cdot \tau \, dx + \int_{\Omega} u \operatorname{div} \tau \, dx = 0.$$

Next, multiply $-\operatorname{div} \sigma = f$ by an arbitrary smooth *scalar-valued* test function v and integrate over Ω :

$$- \int_{\Omega} (\operatorname{div} \sigma) v \, dx = \int_{\Omega} f v \, dx.$$

We leave the divergence on σ rather than integrating the second equation by parts. Now we choose the spaces by requiring every term in these two identities to be well defined. Recall that

$$H(\operatorname{div}; \Omega) = \left\{ \tau \in L^2(\Omega)^d : \operatorname{div} \tau \in L^2(\Omega) \right\}.$$

The term $\int_{\Omega} \sigma \cdot \tau \, dx$ requires both σ and τ to be in $L^2(\Omega)^d$, while $\int_{\Omega} u \operatorname{div} \tau \, dx$ requires $u \in L^2(\Omega)$ and $\operatorname{div} \tau \in L^2(\Omega)$. The second identity additionally requires $\operatorname{div} \sigma \in L^2(\Omega)$ and $v \in L^2(\Omega)$. Thus the natural choices are

$$\sigma, \tau \in H(\operatorname{div}; \Omega), \quad u, v \in L^2(\Omega).$$

Notice that an L^2 function does not have a boundary trace in general, so the condition $u = 0$ is no longer imposed pointwise in the mixed space. It was used in the formal integration-by-parts calculation and is encoded in the first variational identity through the absence of a boundary term. Therefore, the mixed weak problem is: find

$$(\sigma, u) \in H(\operatorname{div}; \Omega) \times L^2(\Omega)$$

such that

$$\begin{cases} \int_{\Omega} \sigma \cdot \tau \, dx + \int_{\Omega} u \operatorname{div} \tau \, dx = 0 & \forall \tau \in H(\operatorname{div}; \Omega), \\ - \int_{\Omega} (\operatorname{div} \sigma) v \, dx = \int_{\Omega} f v \, dx & \forall v \in L^2(\Omega). \end{cases}$$

Equivalently, these two equations can be combined as

$$B((\sigma, u), (\tau, v)) = F(\tau, v)$$

for every $(\tau, v) \in H(\operatorname{div}; \Omega) \times L^2(\Omega)$, with

$$B((\sigma, u), (\tau, v)) = \int_{\Omega} \sigma \cdot \tau \, dx + \int_{\Omega} u \operatorname{div} \tau \, dx - \int_{\Omega} (\operatorname{div} \sigma) v \, dx, \quad F(\tau, v) = \int_{\Omega} f v \, dx.$$

This form is not coercive on the product space. Indeed, for any nonzero $u \in L^2(\Omega)$, take $(\sigma, u) = (0, u)$. Then

$$B((0, u), (0, u)) = 0, \quad \|(0, u)\|_{H(\operatorname{div}) \times L^2} > 0.$$

Consequently, no constant $\alpha > 0$ can satisfy

$$|B(z, z)| \geq \alpha \|z\|^2 \quad \text{for every } z \in H(\operatorname{div}; \Omega) \times L^2(\Omega).$$

Thus the standard Lax–Milgram lemma cannot be applied directly to this mixed formulation. This does not mean that the Poisson problem is ill-posed. Instead, it shows that we need to generalize the Lax–Milgram lemma to cover problems whose bilinear forms are not coercive. In this generalization, coercivity is replaced by an appropriate inf-sup stability condition.

3 Generalized Lax Milgram Lemma and its applications

As shown in Example 6, the mixed formulation of the Poisson equation naturally couples unknowns from two different Hilbert spaces, namely $H(\operatorname{div}; \Omega)$ and $L^2(\Omega)$. In this setting, the associated bilinear form is not coercive on the whole product space, so the classical Lax–Milgram lemma cannot be applied directly. This motivates a generalization of Lax–Milgram to bounded sesquilinear forms $a : X \times Y \rightarrow \mathbb{C}$, where the trial space X and the test space Y may be different Hilbert spaces. The role of coercivity is then replaced by an inf-sup stability condition.

Definition 7 – Inf-sup condition

Let X and Y be normed spaces and let $a : X \times Y \rightarrow \mathbb{C}$ be a bounded sesquilinear form. Its inf-sup constant is

$$\beta = \inf_{u \in X \setminus \{0\}} \sup_{v \in Y \setminus \{0\}} \frac{|a(u, v)|}{\|u\|_X \|v\|_Y}. \quad (3.1)$$

By homogeneity, the inf-sup constant can equivalently be written as

$$\beta = \inf_{\|u\|_X=1} \sup_{\|v\|_Y=1} |a(u, v)|.$$

The condition $\beta > 0$ is also called the Ladyzhenskaya–Babuška–Brezzi (LBB) condition, or inf-sup condition.

Theorem 8 – Generalized Lax–Milgram lemma

Let X and Y be Hilbert spaces and let $a : X \times Y \rightarrow \mathbb{C}$ be bounded. Assume that a satisfies inf-sup condition, i.e., there exists $\alpha > 0$ such that

$$\inf_{\|u\|_X=1} \sup_{\|v\|_Y=1} |a(u, v)| \geq \alpha,$$

and that, for every nonzero $v \in Y$, there exists $u \in X$ with $a(u, v) \neq 0$. Then for every $g \in Y'$ there exists a unique $u \in X$ such that

$$a(u, v) = \overline{g(v)}, \quad v \in Y,$$

and

$$\|u\|_X \leq \frac{1}{\alpha} \|g\|_{Y'}.$$

The proof is almost identical to that of the classical Lax–Milgram lemma. We again use the Riesz representation theorem to construct an operator and then prove that it is bijective. The main difference is that the operator now maps between two possibly different Hilbert spaces, $A : X \rightarrow Y$, and the second assumption of the theorem is needed to establish surjectivity.

Proof. Step 1: Define an operator from X to Y . For each fixed $u \in X$, the map $v \mapsto \overline{a(u, v)}$ is a bounded linear functional on Y . By the Riesz representation theorem, there exists a unique element $Au \in Y$ such that

$$(Au, v) = a(u, v), \quad \forall v \in Y.$$

This defines a bounded linear operator $A : X \rightarrow Y$.

Step 2: Rewrite the original problem as an operator equation. Applying the Riesz representation theorem to $g \in Y'$, there exists a unique $w \in Y$ such that

$$(v, w) = g(v), \quad \forall v \in Y.$$

By conjugate symmetry, $(w, v) = \overline{g(v)}$. Therefore, finding $u \in X$ such that $a(u, v) = \overline{g(v)}$ for every $v \in Y$ is equivalent to solving

$$Au = w.$$

It remains to prove that $A : X \rightarrow Y$ is bijective.

Step 3: Prove that A is injective and has closed range. By the inf-sup condition,

$$\|Au\|_Y = \sup_{\|v\|_Y=1} |(Au, v)| = \sup_{\|v\|_Y=1} |a(u, v)| \geq \alpha \|u\|_X.$$

Thus $Au = 0$ implies $u = 0$, so A is injective. The same lower bound also implies that $\text{Ran } A$ is closed in Y . Indeed, suppose that $Au_n \in \text{Ran } A$ and $Au_n \rightarrow y$ in Y . Then

$$\|u_n - u_m\|_X \leq \frac{1}{\alpha} \|Au_n - Au_m\|_Y.$$

Thus (u_n) is a Cauchy sequence in X . Since X is complete, there exists $u \in X$ such that $u_n \rightarrow u$. The continuity of $A : X \rightarrow Y$ implies that $Au_n \rightarrow Au$. Since $Au_n \rightarrow y$ as well, we obtain $y = Au \in \text{Ran } A$. Therefore, $\text{Ran } A$ is closed in Y .

Step 4: Prove that A is surjective. Suppose $v \in (\text{Ran } A)^\perp$. Then

$$a(u, v) = (Au, v) = 0, \quad \forall u \in X.$$

The second assumption of the theorem implies that $v = 0$. Hence $(\text{Ran } A)^\perp = \{0\}$, and therefore $\text{Ran } A$ is dense in Y . Since $\text{Ran } A$ is both closed and dense, we conclude that $\text{Ran } A = Y$. Thus A is surjective.

Step 5: Conclude existence, uniqueness, and stability. Since $A : X \rightarrow Y$ is bijective, the equation $Au = w$ has a unique solution $u \in X$. Moreover,

$$\alpha \|u\|_X \leq \|Au\|_Y = \|w\|_Y = \|g\|_{Y'},$$

which gives

$$\|u\|_X \leq \frac{1}{\alpha} \|g\|_{Y'}.$$

■

Remark 9 – The operator A is an isomorphism

The proof shows that the induced operator $A : X \rightarrow Y$ is a bounded bijection. Moreover, the inf-sup estimate gives

$$\|A^{-1}y\|_X \leq \frac{1}{\alpha} \|y\|_Y, \quad \forall y \in Y,$$

and hence

$$\|A^{-1}\|_{\mathcal{L}(Y, X)} \leq \frac{1}{\alpha}.$$

Therefore, A is an isomorphism of Hilbert spaces, and the assumptions of the theorem imply that X and Y are topologically isomorphic.

We now turn to mixed variational problems. **From this point onward, all Hilbert spaces and forms are real. In particular, the forms a and b take values in $\mathbb{R}$, not in $\mathbb{C}$.**

Let X and S be real Hilbert spaces, and let

$$a : X \times X \rightarrow \mathbb{R}, \quad b : X \times S \rightarrow \mathbb{R}$$

be bounded bilinear forms. The constraint form b determines the closed subspace

$$Z = \{u \in X : b(u, \xi) = 0 \text{ for every } \xi \in S\}.$$

Definition 10 – Z -Coercivity

The form a is called *coercive on Z* if there exists $\alpha > 0$ such that

$$|a(z, z)| \geq \alpha \|z\|_X^2, \quad \forall z \in Z.$$

Unlike the classical Lax–Milgram lemma, coercivity is required only on the kernel of the constraint, not on the whole space X .

Definition 11 – Babuška–Brezzi condition

The form b is said to satisfy the Babuška–Brezzi condition if there exists $\beta > 0$ such that

$$\sup_{w \in X \setminus \{0\}} \frac{|b(w, p)|}{\|w\|_X} \geq \beta \|p\|_S, \quad \forall p \in S.$$

Equivalently,

$$\inf_{p \in S \setminus \{0\}} \sup_{w \in X \setminus \{0\}} \frac{|b(w, p)|}{\|w\|_X \|p\|_S} \geq \beta > 0.$$

Theorem 12 – Well-posedness of the mixed problem

Let X and S be real Hilbert spaces. Suppose that $a : X \times X \rightarrow \mathbb{R}$ and $b : X \times S \rightarrow \mathbb{R}$ are bounded, that a is coercive on Z , and that b satisfies the Babuška–Brezzi condition. Then, for every $f \in X'$ and $g \in S'$, there exists a unique pair $(u, p) \in X \times S$ such that

$$a(u, \phi) + b(\phi, p) = f(\phi), \quad \forall \phi \in X,$$

and

$$b(u, \xi) = g(\xi), \quad \forall \xi \in S.$$

Moreover, there exists a constant $C > 0$, depending only on the continuity, kernel-coercivity, and inf-sup constants, such that

$$\|u\|_X + \|p\|_S \leq C (\|f\|_{X'} + \|g\|_{S'}).$$

Proof. Step 1: Show that the second equation always has a solution. The second equation involves only the form b , so we first rewrite b as an operator. For each $\phi \in X$, the Riesz representation theorem gives a unique $B\phi \in S$ such that

$$b(\phi, \xi) = (B\phi, \xi)_S = (\phi, B^*\xi)_X, \quad \forall \xi \in S,$$

where $B : X \rightarrow S$ is bounded and $B^* : S \rightarrow X$ is its adjoint. The LBB condition now says exactly that B^* is bounded from below:

$$\|B^*\xi\|_X = \sup_{\phi \in X \setminus \{0\}} \frac{|b(\phi, \xi)|}{\|\phi\|_X} \geq \beta \|\xi\|_S, \quad \forall \xi \in S.$$

To prove that B is surjective, it is useful to look at $BB^* : S \rightarrow S$. Indeed, if $BB^*\eta = q$, then $q = B(B^*\eta)$ already belongs to the range of B . The LBB estimate gives

$$(BB^*\xi, \xi)_S = \|B^*\xi\|_X^2 \geq \beta^2 \|\xi\|_S^2, \quad \forall \xi \in S.$$

Therefore the bilinear form $c(\eta, \xi) = (B^*\eta, B^*\xi)_X$ is bounded and coercive on S . The Lax–Milgram lemma implies that, for every $g \in S'$, there is a unique $\eta \in S$ such that

$$(BB^*\eta, \xi)_S = g(\xi), \quad \forall \xi \in S.$$

Thus BB^* is bijective, and since $\text{Ran}(BB^*) \subseteq \text{Ran } B$, the operator B is surjective. This proves that the constraint equation $b(u, \xi) = g(\xi)$ always has a solution. More precisely, if η is the solution above and we set $u_0 = B^*\eta$, then $Bu_0 = BB^*\eta$, so u_0 solves the constraint equation. Moreover,

$$\|u_0\|_X^2 = (BB^*\eta, \eta)_S = g(\eta) \leq \|g\|_{S'} \|\eta\|_S, \quad \beta \|\eta\|_S \leq \|B^*\eta\|_X = \|u_0\|_X.$$

Combining these inequalities gives the useful estimate

$$\|u_0\|_X \leq \frac{1}{\beta} \|g\|_{S'}.$$

Step 2: Correct a constraint solution inside the kernel Z . Choose the controlled solution u_0 constructed in Step 1. Any other solution must differ from u_0 by an element of Z , because $b(u - u_0, \xi) = 0$ for every $\xi \in S$. It is therefore natural to look for the desired solution in the form $u = u_0 + u_1$, where $u_1 \in Z$.

Before applying Lax–Milgram, we must check that Z is itself a Hilbert space. Since $Z = \text{Ker } B$ and B is continuous, Z is a closed subspace of X . Hence the inner product of X , restricted to Z , makes Z a Hilbert space.

We now test the first equation only with $\phi \in Z$. The multiplier term vanishes because $b(\phi, p) = 0$, so the equation that determines u_1 is

$$a(u_1, \phi) = f(\phi) - a(u_0, \phi), \quad \forall \phi \in Z.$$

The right-hand side is a bounded linear functional on Z , while the restriction of a to $Z \times Z$ is bounded and coercive. The Lax–Milgram lemma therefore gives a unique $u_1 \in Z$. If C_a denotes the continuity constant of a , then

$$\|u_1\|_X \leq \frac{1}{\alpha} (\|f\|_{X'} + C_a \|u_0\|_X).$$

Step 3: Recover the multiplier p . At this point we have found $u_1 \in Z$, and hence $u = u_0 + u_1$. Because $u_1 \in Z$, this function still satisfies the second equation automatically. It remains to choose $p \in S$ so that

$$a(u, \phi) + b(\phi, p) = f(\phi), \quad \forall \phi \in X.$$

Since Z is closed, the Hilbert space X has the orthogonal decomposition $X = Z \oplus Z^\perp$. Step 2 already guarantees the first equation for every $\phi \in Z$. Therefore, it is enough to find p such that

$$b(\phi, p) = \ell(\phi), \quad \forall \phi \in Z^\perp, \quad \ell(\phi) := f(\phi) - a(u, \phi).$$

The functional ℓ is bounded on $Z^\perp$.

We now apply the generalized Lax–Milgram lemma to the bounded bilinear form $\tilde{b} : S \times Z^\perp \rightarrow \mathbb{R}$ defined by $\tilde{b}(p, \phi) = b(\phi, p)$. We first explain carefully why the supremum in the LBB condition is unchanged when the test functions are restricted from X to $Z^\perp$.

Fix $p \in S$. Every $w \in X$ has a unique orthogonal decomposition $w = z + \phi$, where $z \in Z$ and $\phi \in Z^\perp$. Since $b(z, p) = 0$, we have $b(w, p) = b(\phi, p)$. Moreover, orthogonality gives $\|\phi\|_X \leq \|w\|_X$. Therefore, if $\phi \neq 0$,

$$\frac{|b(w, p)|}{\|w\|_X} = \frac{|b(\phi, p)|}{\|w\|_X} \leq \frac{|b(\phi, p)|}{\|\phi\|_X}.$$

If $\phi = 0$, then $w \in Z$ and $b(w, p) = 0$, so the same conclusion is trivial. Taking the supremum over $w \in X \setminus \{0\}$ consequently gives

$$\sup_{w \in X \setminus \{0\}} \frac{|b(w, p)|}{\|w\|_X} \leq \sup_{\phi \in Z^\perp \setminus \{0\}} \frac{|b(\phi, p)|}{\|\phi\|_X}.$$

The reverse inequality follows immediately from $Z^\perp \subset X$. Hence the two suprema are equal:

$$\sup_{w \in X \setminus \{0\}} \frac{|b(w, p)|}{\|w\|_X} = \sup_{\phi \in Z^\perp \setminus \{0\}} \frac{|b(\phi, p)|}{\|\phi\|_X}.$$

The original LBB condition therefore implies

$$\inf_{p \in S \setminus \{0\}} \sup_{\phi \in Z^\perp \setminus \{0\}} \frac{|\tilde{b}(p, \phi)|}{\|p\|_S \|\phi\|_X} \geq \beta > 0.$$

We must also check nondegeneracy in the second variable. If $0 \neq \phi \in Z^\perp$, then $\phi \notin Z = \text{Ker } B$, so $B\phi \neq 0$. Choosing $p = B\phi \in S$ gives

$$\tilde{b}(p, \phi) = b(\phi, B\phi) = \|B\phi\|_S^2 > 0.$$

Both hypotheses of the generalized Lax–Milgram lemma are therefore satisfied. Consequently, there exists a unique $p \in S$ such that $b(\phi, p) = \ell(\phi)$ for every $\phi \in Z^\perp$. Together with the identity already established on Z , this proves the first equation for every $\phi \in X$. Hence (u, p) solves the mixed problem.

Step 4: Prove uniqueness. Because the mixed problem is linear, it is enough to show that the homogeneous problem has only the zero solution. Suppose therefore that $f = 0$ and $g = 0$, and let $(u, p) \in X \times S$ be a solution. The second equation gives $b(u, \xi) = 0$ for every $\xi \in S$, so $u \in Z$. Taking $\phi = u$ in the first equation and $\xi = p$ in the second equation gives, respectively,

$$a(u, u) + b(u, p) = 0, \quad b(u, p) = 0.$$

Subtracting the second identity from the first yields $a(u, u) = 0$. Since $u \in Z$, the Z -coercivity of a implies

$$\alpha \|u\|_X^2 \leq |a(u, u)| = 0,$$

and hence $u = 0$. The first equation now reduces to $b(\phi, p) = 0$ for every $\phi \in X$. Finally, the LBB condition gives

$$\beta \|p\|_S \leq \sup_{\phi \in X \setminus \{0\}} \frac{|b(\phi, p)|}{\|\phi\|_X} = 0,$$

so $p = 0$. Thus the homogeneous problem has only the zero solution, and the solution constructed above is unique.

Step 5: Derive the stability estimate. The estimates from Steps 1 and 2 first give control of u . Since $u = u_0 + u_1$,

$$\begin{aligned} \|u\|_X &\leq \|u_0\|_X + \|u_1\|_X \\ &\leq \frac{1}{\alpha} \|f\|_{X'} + \frac{1}{\beta} \left(1 + \frac{C_a}{\alpha}\right) \|g\|_{S'}. \end{aligned}$$

Thus u depends continuously on the data f and g . To estimate p , use the first mixed equation together with the LBB condition. For every $\phi \in X$, $b(\phi, p) = f(\phi) - a(u, \phi)$, and therefore

$$\begin{aligned} \beta \|p\|_S &\leq \sup_{\phi \in X \setminus \{0\}} \frac{|b(\phi, p)|}{\|\phi\|_X} \\ &\leq \|f\|_{X'} + C_a \|u\|_X. \end{aligned}$$

Substituting the estimate for u into this inequality shows that $\|p\|_S$ is also bounded by a constant times $\|f\|_{X'} + \|g\|_{S'}$. Consequently, there exists a constant $C > 0$, depending only on α , β , and the continuity constant C_a , such that

$$\|u\|_X + \|p\|_S \leq C (\|f\|_{X'} + \|g\|_{S'}).$$

■

4 A Discrete version of Everything Above

In the preceding sections, we developed the well-posedness theory for continuous variational problems. The Lax–Milgram lemma, its generalized version, and the LBB condition provided conditions under which a variational problem admits a unique solution that depends continuously on the given data. In finite element methods, however, the continuous trial and test spaces are replaced by

finite-dimensional subspaces. We must therefore understand which parts of the continuous theory remain valid after discretization and whether the resulting stability estimates are uniform with respect to the mesh size. The discrete results introduced below form the foundation of finite element error analysis: they connect stability of the discrete problem with quasi-optimal approximation and, ultimately, with convergence of the finite element solution to the exact solution.

4.1 Céa's Lemma

We begin with the conforming Galerkin approximation of a coercive variational problem. In this setting, Céa's lemma shows that the discretization error is controlled, up to a stability constant, by the best approximation error in the chosen finite-dimensional space. It is the basic link between well-posedness of the continuous problem and convergence of its finite element approximation.

Lemma 13 – Céa's Lemma

Let X be a Hilbert space and let $X_h \subset X$ be a finite-dimensional subspace. Suppose that $a : X \times X \rightarrow \mathbb{C}$ is a bounded coercive sesquilinear form: there exist constants $M, \alpha > 0$ such that

$$|a(w, v)| \leq M \|w\|_X \|v\|_X, \quad |a(v, v)| \geq \alpha \|v\|_X^2 \quad \forall w, v \in X.$$

For $f \in X'$, let $u \in X$ be the continuous solution satisfying

$$a(u, v) = \overline{f(v)}, \quad \forall v \in X.$$

Then the discrete problem of finding $u_h \in X_h$ such that

$$a(u_h, v_h) = \overline{f(v_h)}, \quad \forall v_h \in X_h,$$

has a unique solution. Moreover,

$$\|u - u_h\|_X \leq \frac{M}{\alpha} \inf_{w_h \in X_h} \|u - w_h\|_X.$$

The constants M and α , and hence the factor M/α , are independent of h .

Proof. Because X_h is finite-dimensional, it is a closed subspace of X and is therefore itself a Hilbert space. The restrictions of a and f to X_h retain the same boundedness and coercivity estimates. The Lax–Milgram lemma applied on X_h consequently gives a unique discrete solution $u_h \in X_h$.

The central observation is the *Galerkin orthogonality*. Testing the continuous problem with an arbitrary $v_h \in X_h$ and subtracting the discrete equation gives

$$a(u - u_h, v_h) = 0, \quad \forall v_h \in X_h.$$

Thus the error is orthogonal to the discrete space in the sense induced by the form a , even when a is not an inner product.

Let $w_h \in X_h$ be arbitrary and set $e = u - u_h$. Since $w_h - u_h \in X_h$, Galerkin orthogonality yields

$$a(e, e) = a(e, u - w_h) + a(e, w_h - u_h) = a(e, u - w_h).$$

Using coercivity and boundedness, we obtain

$$\alpha \|e\|_X^2 \leq |a(e, e)| = |a(e, u - w_h)| \leq M \|e\|_X \|u - w_h\|_X.$$

If $e = 0$, the result is immediate. Otherwise, dividing by $\alpha \|e\|_X$ gives

$$\|u - u_h\|_X \leq \frac{M}{\alpha} \|u - w_h\|_X.$$

Since $w_h \in X_h$ was arbitrary, taking the infimum over X_h proves the estimate. ■

Remark 14 – Quasi-optimality

Céa's estimate is called *quasi-optimal*: the actual Galerkin error is, up to the fixed factor M/α , no larger than the best approximation error available in X_h . Thus convergence reduces to an approximation question. If

$$\inf_{w_h \in X_h} \|u - w_h\|_X \longrightarrow 0 \quad \text{as } h \rightarrow 0,$$

then $u_h \rightarrow u$ in X . A best approximant may exist without being the Galerkin solution; Céa's lemma says that the Galerkin solution is nevertheless uniformly comparable to it.

4.2 Discrete Mixed Problem

Let $X_h \subset X$ and $S_h \subset S$ be finite-dimensional subspaces, where the parameter $h > 0$ usually represents the mesh size. The subscript h indicates that both spaces, and hence the numerical solution, may change as the mesh is refined.

Definition 15 – Discrete mixed problem

Given $f \in X'$ and $g \in S'$, the discrete mixed problem is:

- (i) find a discrete primal variable $u_h \in X_h$ and a discrete multiplier $p_h \in S_h$;
- (ii) require the first equation

$$a(u_h, \phi_h) + b(\phi_h, p_h) = f(\phi_h), \quad \forall \phi_h \in X_h;$$

- (iii) require the discrete constraint

$$b(u_h, \xi_h) = g(\xi_h), \quad \forall \xi_h \in S_h.$$

This is the natural Galerkin approximation of the continuous mixed problem: the equations are unchanged, but trial and test functions are restricted to the finite-dimensional spaces.

Definition 16 – Discrete constraint spaces

The discrete analogues of the kernel and the affine constraint set are defined as follows.

- (i) The *discrete kernel* is

$$Z_h = \{v_h \in X_h : b(v_h, \xi_h) = 0 \text{ for every } \xi_h \in S_h\}.$$

- (ii) For a given $g \in S'$, the *discrete affine constraint set* is

$$Z_h(g) = \{v_h \in X_h : b(v_h, \xi_h) = g(\xi_h) \text{ for every } \xi_h \in S_h\}.$$

Thus $Z_h = Z_h(0)$, while every function in $Z_h(g)$ satisfies the second equation of the discrete mixed problem.

Remark 17 – The discrete kernel may be larger

In general, Z_h need not equal $X_h \cap Z$. Membership in Z_h tests the constraint only against functions in S_h , whereas membership in the continuous kernel Z requires the constraint to hold against every function in S . This distinction is one reason why stability must be verified again at the discrete level.

Definition 18 – Uniform discrete stability

The family of discrete spaces (X_h, S_h) is stable when the following two properties hold with constants independent of h .

- (i) The form a is *uniformly coercive on the discrete kernel*: there exists $\alpha > 0$, independent of h , such that

$$|a(v_h, v_h)| \geq \alpha \|v_h\|_X^2, \quad \forall v_h \in Z_h.$$

- (ii) The form b satisfies the *discrete LBB condition*: there exists $\beta > 0$, independent of h , such that

$$\sup_{\phi_h \in X_h \setminus \{0\}} \frac{|b(\phi_h, q_h)|}{\|\phi_h\|_X} \geq \beta \|q_h\|_S, \quad \forall q_h \in S_h.$$

Equivalently,

$$\inf_{q_h \in S_h \setminus \{0\}} \sup_{\phi_h \in X_h \setminus \{0\}} \frac{|b(\phi_h, q_h)|}{\|\phi_h\|_X \|q_h\|_S} \geq \beta > 0.$$

The independence of α and β from h is essential. For a fixed finite-dimensional pair, existence may still hold when these constants depend on h , but deterioration of the constants as $h \rightarrow 0$ destroys the uniform estimates needed for finite element convergence.

5 Sobolev spaces

The treatment of Sobolev spaces in Evans' PDE textbook is concise and highly effective for many standard applications, but it does not include all of the results needed later in these notes, and some of the proofs are technically lengthy. Peter Monk's presentation contains a broader collection of relevant results, especially for finite element analysis and spaces arising in electromagnetic problems, but its main structure of the theory is difficult to understand and remember at a first reading. In this section, we reorganize and restate the most important results from Chapter 3 of Monk's book. The aim is to provide a natural extension of the discussion of Sobolev spaces in Chapter 5 of Evans, while emphasizing the connections among the definitions, trace results, density theorems, and differential-operator spaces. The material is arranged so that the principal statements and their roles are easier to understand, compare, and remember.

We begin with the basic Sobolev spaces. Let $\Omega \subset \mathbb{R}^d$ be an open set, and let $\alpha = (\alpha_1, \dots, \alpha_d)$ be a multi-index. We write

$$|\alpha| = \alpha_1 + \dots + \alpha_d, \quad D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

For a nonsmooth function, $D^\alpha u$ is always understood in the distributional sense. We assume that the reader is already familiar with distributions and distributional derivatives; readers who need a review may consult the relevant Wikipedia article or any standard textbook on functional analysis or partial differential equations.

Definition 19 – Sobolev space $W^{k,p}(\Omega)$

Let $k \in \mathbb{N}_0$ and $1 \leq p \leq \infty$. The Sobolev space $W^{k,p}(\Omega)$ consists of all functions $u \in L^p(\Omega)$ whose distributional derivatives up to order k are represented by functions in $L^p(\Omega)$:

$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for every } |\alpha| \leq k\}.$$

For $1 \leq p < \infty$, this space is equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p},$$

while for $p = \infty$, we use

$$\|u\|_{W^{k,\infty}(\Omega)} = \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}.$$

With these norms, $W^{k,p}(\Omega)$ is a Banach space.

The case $k = 0$ gives $W^{0,p}(\Omega) = L^p(\Omega)$. When $p = 2$, the Sobolev space is a Hilbert space, and we use the standard notation

$$H^k(\Omega) = W^{k,2}(\Omega), \quad (u, v)_{H^k(\Omega)} = \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}.$$

5.1 Sobolev Embeddings

Sobolev embedding theorems describe how integrability of a function improves when sufficiently many of its distributional derivatives are integrable.

Domain assumption: Throughout this subsection, Ω is either the whole space $\mathbb{R}^n$, or a bounded open subset of $\mathbb{R}^n$ with Lipschitz boundary.

In the second case, the Lipschitz boundary assumption means that the boundary is locally the graph of a Lipschitz function. More precisely, near every point of $\partial\Omega$, one can translate and rotate the coordinates so that, in a neighborhood of that point,

$$\partial\Omega = \{(x', x_n) : x_n = \psi(x')\},$$

where ψ is Lipschitz continuous, and Ω lies locally on one side of this graph. Thus the boundary may have corners, but it cannot have cusps that are sharper than a Lipschitz graph.

The Sobolev embedding is governed by the single quantity

$$\frac{1}{p^*} = \frac{1}{p} - \frac{k}{n}.$$

Here and below, we use the convention $1/\infty = 0$. The notation p^* represents an actual Lebesgue exponent only when the right-hand side is positive. In the other two cases, only its sign matters.

Theorem 20 – Sobolev embedding theorem

Let $1 \leq p \leq \infty$, let $k \geq 1$ be an integer, and suppose that Ω is either $\mathbb{R}^n$ or a bounded open subset of $\mathbb{R}^n$ with Lipschitz boundary. The principal Sobolev embeddings are summarized below, where $\hookrightarrow$ denotes a continuous embedding.

Case	Embedding	Range or parameters
$\frac{1}{p^*} > 0$	$W^{k,p}(\Omega) \hookrightarrow L^q(\Omega)$	$p \leq q \leq p^*$.
$\frac{1}{p^*} = 0$	$W^{k,p}(\Omega) \hookrightarrow L^q(\Omega)$	$p \leq q < \infty$.
$\frac{1}{p^*} < 0$	$W^{k,p}(\Omega) \hookrightarrow C^{m,\gamma}(\overline{\Omega})$	Write $s = k - n/p > 0$. If $s = m + \gamma$, take $m \in \mathbb{N}_0$ and $0 < \gamma < 1$. If s is an integer, take $m = s - 1$ and any $0 < \gamma < 1$. For $p = \infty$, one has the stronger embedding $W^{k,\infty}(\Omega) \hookrightarrow C^{k-1,1}(\overline{\Omega})$.

The preceding theorem gives continuous embeddings. On a bounded domain, one can obtain the stronger conclusion that bounded sequences in the Sobolev space have convergent subsequences in a slightly weaker target space. This is the content of the Rellich–Kondrachov theorem. We write $\Subset$ for a compact embedding.

Theorem 21 – Rellich–Kondrachov compact embedding theorem

Let $1 \leq p \leq \infty$, let $k \geq 1$ be an integer, and let Ω be a **bounded** open subset of $\mathbb{R}^n$ with Lipschitz boundary. The continuous Sobolev embeddings and their compact counterparts can be compared as follows.

The boundedness assumption is essential: unlike the continuous Sobolev embedding theorem, this compact embedding theorem does not hold on the whole space $\mathbb{R}^n$ in general.

Sobolev continuous embedding	Rellich–Kondrachov compact embedding
If $\frac{1}{p^*} > 0$, then $W^{k,p}(\Omega) \hookrightarrow L^q(\Omega)$ for $p \leq q \leq p^*$.	If $\frac{1}{p^*} > 0$, then $W^{k,p}(\Omega) \Subset L^q(\Omega)$ for $1 \leq q < p^*$.
If $\frac{1}{p^*} = 0$, then $W^{k,p}(\Omega) \hookrightarrow L^q(\Omega)$ for $p \leq q < \infty$.	If $\frac{1}{p^*} = 0$, then $W^{k,p}(\Omega) \Subset L^q(\Omega)$ for $1 \leq q < \infty$.
If $\frac{1}{p^*} < 0$, write $s = k - n/p > 0$. When $s = m + \gamma$ with $0 < \gamma < 1$, $W^{k,p}(\Omega) \hookrightarrow C^{m,\gamma}(\overline{\Omega})$.	Under the same assumptions, $W^{k,p}(\Omega) \Subset C^{m,\gamma'}(\overline{\Omega})$ for every $0 < \gamma' < \gamma$.

In the Morrey-type case where s is an integer, the continuous embedding holds into $C^{s-1,\gamma}(\overline{\Omega})$ for every $0 < \gamma < 1$, and the embedding into each such space is compact. For $p = \infty$, one has

$$W^{k,\infty}(\Omega) \hookrightarrow C^{k-1,1}(\overline{\Omega}), \quad W^{k,\infty}(\Omega) \Subset C^{k-1,\gamma}(\overline{\Omega}) \quad (0 \leq \gamma < 1).$$

The boundedness of Ω is essential for compactness. On $\mathbb{R}^n$, a fixed function may be translated without changing its Sobolev norm, and the resulting sequence need not contain a convergent subsequence in the target space. See the following example.

Example 22 – Translations destroy compactness on $\mathbb{R}^n$

Choose a nonzero function $\varphi \in C_c^\infty(\mathbb{R}^n)$, and suppose for simplicity that $\text{supp } \varphi \subset B(0, R)$ for some $R > 0$. Let $e_1 = (1, 0, \dots, 0)$ and define

$$\varphi_j(x) = \varphi(x - 3Rje_1), \quad j = 1, 2, \dots$$

Translation does not change any L^p -norm of a derivative, because $D^\alpha \varphi_j(x) = D^\alpha \varphi(x - 3Rje_1)$. Consequently,

$$\|\varphi_j\|_{W^{k,p}(\mathbb{R}^n)} = \|\varphi\|_{W^{k,p}(\mathbb{R}^n)} \quad \text{for every } j,$$

so (φ_j) is a bounded sequence in every Sobolev space to which φ belongs. On the other hand, the supports of the functions φ_j are pairwise disjoint. Hence, whenever $j \neq \ell$ and $1 \leq q < \infty$,

$$\|\varphi_j - \varphi_\ell\|_{L^q(\mathbb{R}^n)}^q = \|\varphi_j\|_{L^q(\mathbb{R}^n)}^q + \|\varphi_\ell\|_{L^q(\mathbb{R}^n)}^q = 2 \|\varphi\|_{L^q(\mathbb{R}^n)}^q.$$

Thus any two distinct members of the sequence have the same positive distance $2^{1/q} \|\varphi\|_{L^q(\mathbb{R}^n)}$. No subsequence can therefore be Cauchy, and hence no subsequence can converge in $L^q(\mathbb{R}^n)$. This example shows concretely why the corresponding Sobolev embedding on the whole space is not compact: a bounded family can escape to infinity by translation.

5.2 Embedding on Lower-Dimensional Sections

Sobolev regularity also allows a function to be restricted to a lower-dimensional set, provided that the function has enough smoothness to compensate for the loss of dimension. Let $P \subset \mathbb{R}^n$ be an affine hyperplane of dimension ℓ , where $1 \leq \ell \leq n$, and set

$$\Omega' = \Omega \cap P.$$

We regard Ω' as an ℓ -dimensional domain with the measure and derivatives induced by P . When $\ell < n$, the notation $u|_{\Omega'}$ must be interpreted as a trace rather than as the pointwise restriction of an arbitrary Sobolev function. For a smooth function the restriction is classical, and the estimates below show that this restriction operator extends continuously to the indicated Sobolev space. Equivalently, if smooth functions u_r converge to u in the source space, then $u_r|_{\Omega'}$ converges in the stated target space, and its limit defines the trace of u on Ω' .

For an ℓ -dimensional section, it is convenient to introduce the rescaled critical exponent p_ℓ^* by

$$\frac{1}{p_\ell^*} = \frac{n}{\ell} \left(\frac{1}{p} - \frac{k}{n} \right).$$

As before, when the right-hand side is nonpositive, only its sign is relevant.

Theorem 23 – Trace estimates on lower-dimensional sections

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary, let $k \in \mathbb{N}_0$, let $1 \leq p \leq \infty$, and let $\Omega' = \Omega \cap P$, where P is an ℓ -dimensional affine hyperplane. Whenever $\ell < n$, the restriction in the table is understood as the trace described above. In this theorem, p_ℓ^* is defined by the preceding formula.

Case	Bounded map	Range or parameters
$0 < \frac{1}{p_\ell^*} < \frac{1}{p}$	$\text{Tr}_{\Omega'} : W^{k,p}(\Omega) \rightarrow L^q(\Omega')$	$p \leq q \leq p_\ell^*$.
$\frac{1}{p_\ell^*} = 0$	$\text{Tr}_{\Omega'} : W^{k,p}(\Omega) \rightarrow L^q(\Omega')$	$1 \leq \ell \leq n, \quad p \leq q < \infty$.
$\frac{1}{p_\ell^*} < 0$	$W^{k,p}(\Omega) \hookrightarrow C^{m,\alpha}(\overline{\Omega})$	Write $s = k - n/p > 0$. If $s = m + \alpha$, take $m \in \mathbb{N}_0$ and $0 < \alpha < 1$. If s is an integer, take $m = s - 1$ and any $0 < \alpha < 1$.

In the first two rows, boundedness means that there exists a constant $C > 0$ such that

$$\|\text{Tr}_{\Omega'} u\|_{L^q(\Omega')} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

Remark 24 – What the theorem says about the trace

An element of $W^{k,p}(\Omega)$ is defined only up to a set of n -dimensional measure zero. When $\ell < n$, the section Ω' itself has n -dimensional measure zero. We may therefore change an arbitrary representative of u on all of Ω' without changing the element of $W^{k,p}(\Omega)$. This is why the expression $u|_{\Omega'}$ cannot be defined by simply evaluating an arbitrary representative. The trace is instead defined by approximation. For a smooth function v , restriction is unambiguous, and we set

$$\text{Tr}_{\Omega'} v = v|_{\Omega'}.$$

Here we use the fact that $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$ for $1 \leq p < \infty$. Thus, for any $u \in W^{k,p}(\Omega)$, we may choose a sequence $(u_r) \subset C^\infty(\overline{\Omega})$ such that

$$u_r \longrightarrow u \quad \text{in } W^{k,p}(\Omega).$$

Under the condition

$$0 \leq \frac{1}{p_\ell^*} < \frac{1}{p},$$

the estimate in the theorem gives

$$\|u_r|_{\Omega'} - u_s|_{\Omega'}\|_{L^q(\Omega')} \leq C \|u_r - u_s\|_{W^{k,p}(\Omega)}.$$

Thus the restrictions $(u_r|_{\Omega'})$ form a Cauchy sequence in $L^q(\Omega')$. We define the trace of u by

$$\text{Tr}_{\Omega'} u := \lim_{r \rightarrow \infty} u_r|_{\Omega'} \quad \text{in } L^q(\Omega').$$

The same estimate shows that the limit is independent of the chosen smooth approximating sequence. Hence this construction gives a well-defined bounded linear operator

$$\text{Tr}_{\Omega'} : W^{k,p}(\Omega) \longrightarrow L^q(\Omega').$$

The table specifies which values of q are allowed. In the Morrey-type case, every Sobolev function already has a continuous representative, so its trace is simply the ordinary pointwise restriction of that representative.